

# Arduino Based Wireless Visitors Counter

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Information about the total number of persons present inside a building such as a hospital, shopping mall, or factory helps provide crucial information to the security/safety personnel during emergency situations. This article presents the implementation of an Arduino based wireless visitors counter. The authors' prototypes of the transmitter and receiver are shown in Fig. 1 and Fig. 2, respectively while their block diagrams are shown in Fig. 3 and Fig. 4.

## Circuit and working

The main components used in this project are:

**Arduino Uno board.** Arduino Uno is an AVR ATmega328P microcontroller based development board with six analogue input pins and 14 digital I/O pins. The microcontroller has 32kB ISP flash memory, 2kB RAM, and 1kB EEPROM and can operate at a clock frequency of 16MHz. The Arduino Uno board (Board1) provides capability of serial communication via UART, SPI, and I2C. In this project, the board is used for the receiver. Digital I/O pins 12, 11, 7, 6, 5, and 4 of Board1 are used to interface with the LCD, and its digital I/O pins 2 and 3 are configured as interrupt pins.

**HT12E.** The  $2^{12}$  encoders are a series of CMOS LSIs generally used for remote control system applications. These are capable of encoding information consisting of N address bits and 12-N data bits. Each address/data input can be set to one of the two logic states. The programmed address/data are transmitted together with the header bits via an RF or an infrared transmission medium upon receipt of a trigger signal.

**HT12D.** The  $2^{12}$  decoders are a series of CMOS LSIs that are paired with HT12E encoders for use in remote control system applications.

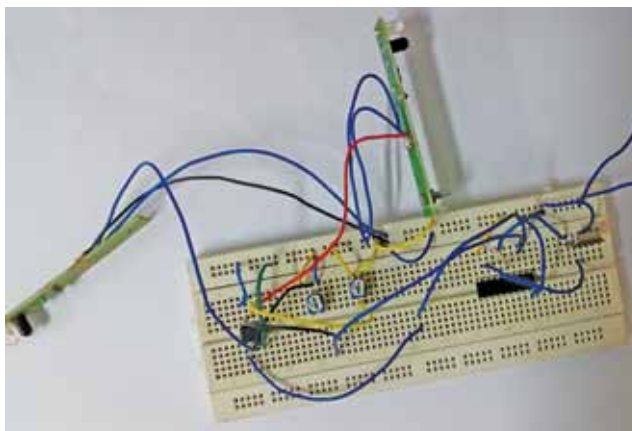


Fig. 1: Authors' prototype of the transmitter

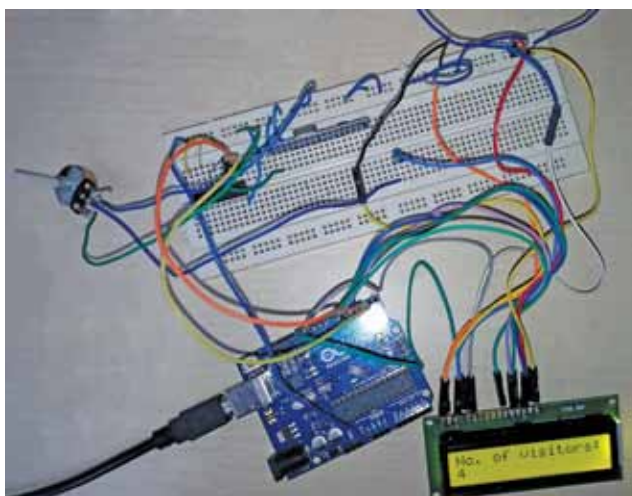


Fig. 2: Authors' prototype of the receiver

The decoders receive serial addresses and data from a programmed  $2^{12}$  series of encoders that are transmitted by a carrier using an RF or an IR transmission medium. These compare the serial input data three times continuously with their local addresses. If no error or unmatched codes are found, the input data codes are decoded and transferred to the output pins. The  $2^{12}$  decoders are capable of decoding information that consists of N bits of addresses and 12-N bits of data. For a

valid transmission, the address bits of both encoder and decoder must be same.

**16 × 2 character LCD.** The LCD is used to display the total number of visitors entering the building. The LCD has sixteen columns and two rows for displaying data. The backlit LCD used here is interfaced with the Arduino in a 4-bit mode. The LCD pins RS, EN, D4, D5, D6, D7 are connected to digital I/O pins 12, 11, 7, 6, 5, 4 of the Arduino Uno, respectively, on the receiver.

**434MHz modules.** The 434MHz transmitter (TX) and receiver (RX) modules are electronic devices used to transmit and receive radio frequency signals

between two devices. These modules are used widely in electronic design to avoid the complexity in designing radio circuitry. The carrier frequency of the modules used here is 434MHz.

**LM358.** This IC has of two independent, high gain, and internally frequency compensated operational amplifiers which are designed specifically to operate from a single power supply over a wide range of voltages. Operation from split power supplies is also possible. In this project the